Divergence of Fear Gauges and Stock Market Returns

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Abstract

This paper investigates the Divergence of Fear Gauges (DFG) between the Merrill Lynch Option Volatility Estimate (MOVE) Index and the Chicago Board Options Exchange Volatility Index (VIX) as a predictor of future equity market returns. We apply an equilibrium model to theoretically examine the relationship between DFG and stock market returns. The model suggests that DFG could negatively predict future equity market returns. Utilizing a dataset from 2004 to 2022, we empirically test the predictive effects of DFG and document results consistent with the model's implication. The predictive effects of DFG on equity returns remain consistent across various robustness tests, including those adopting alternative measures, accounting for the "flight to safety" effect, controlling for risk aversion, and across different forecast intervals, both in-sample and out-of-sample.

Keywords: Fear gauges, divergence, volatility, MOVE, VIX, market returns

JEL classification: G12, G14

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1. Introduction

Following the implementation of quantitative easing in response to the COVID-19 pandemic, market participants have intensely scrutinized the Federal Reserve's monetary policy, particularly its adjustments to the federal funds rate amid varying inflation levels. This heightened attention has brought interest rate volatility that is closely related to uncertainty about the path of the federal funds rate (Sarisoy, 2023), to the forefront of financial analysis. Among various measures of interest rate volatility, the Merrill Lynch Option Volatility Estimate (MOVE) Index is one of the most prominent proxies.[1] The MOVE Index, often referred to as the "fear gauge" of the Treasury bond market, measures the expected volatility in Treasury yields based on a basket of Treasury bond options. It operates similarly to the Chicago Board Options Exchange Volatility Index (VIX), which serves as the fear gauge for the equity market. Whether the MOVE Index serves as a reliable indicator for future rate changes by the Federal Reserve remains debated among practitioners (e.g., Bartolini et al., 2019; Hindlian and Bassma, 2023).

While both Treasury bond and equity markets react sensitively to news about changes in the Federal Reserve's interest rates and other common shocks (e.g., Connolly, Stivers and Sun, 2005; Baele, Bekaert and Inghelbrecht, 2010; Hartmann, Straetmans and de Vries, 2004; Guidolin and Timmermann, 2006; Gürkaynak, Sack and Swanson, 2005; Greenwood, Hanson and Stein, 2020) as captured by the MOVE and VIX indices, there are periods characterized by a divergence of these fear gauges. Such divergence has been reported by reputable financial media outlets such as *the Wall Street Journal* and *Financial Times* (e.g., Georgiadis, 2019; Banerji, 2019; Steer, 2023), suggesting it might signal future market fluctuations. Although

[1] The MOVE index was introduced in 1988. It was acquired by Intercontinental Exchange in 2019 and now referred as ICE BofAML MOVE Index. The VIX index was officially launched in 1993 with historical data stretching back to 1990 as a result of retroactive calculations based on historical options pricing data.

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extensive studies have explored these indices separately, analyzing their properties and relationships with other financial variables (e.g., Whaley, 2000; Andersen, Bondarenko and Gonzalez-Perez, 2015; Bollerslev, Tauchen and Zhou, 2009; Fleming, Kirby and Ostdiek, 1998), their divergence has been rarely investigated.

Motivated by these, this paper aims to fill this gap in the literature and investigate the divergence of fear gauges (DFG), particularly examining if DFG predicts future equity market returns as argued by leading financial media. We define DFG as the relative distance between the implied Treasury rate (bond market) volatility and implied equity market volatility. Specifically, a high (low) DFG indicates that the interest rate volatility is relatively high (low) compared to the equity market volatility. To estimate the potential predictive effect of DFG on equity market returns, we adopt and extend the equilibrium model from Bollerslev et al. (2009) to theoretically investigate the relationship between DFG and future equity market returns. The extended model suggests that DFG, similar to the variance risk premium (VRP) in Bollerslev et al. (2009), should serve as a useful predictor, with volatility-of-volatility constituting the dominant source of variation in both the DFG and the equity premium.

To empirically test the predictability, we apply the residuals of regressing MOVE on VIX as a primary divergence measure.[2] Consistent with our definition, a positive (negative) DFG indicates that MOVE is relatively greater (smaller) than VIX. Employing a dataset

[2] We use the MOVE index as the primary proxy for interest rate volatility for two main reasons. First, the MOVE index is well-established and widely used by both practitioners (e.g., Eisen, 2017; Otani, 2018; Frame and McCormick, 2023) and in high-standard academic studies (e.g., Carlin et al., 2014; Di Maggio et al., 2017; Moench and Soofi-Siavash, 2022) to measure Treasury bond market volatility. More importantly, the MOVE index is timely and publicly accessible, facilitating the investigation of its divergence from the VIX to provide better practical implications for market participants. While some may contend that the MOVE index relies on the Treasury options market, which is perceived to be less liquid than the Treasury futures options market, it is important to note that alternative interest rate volatility indices based on Treasury futures options—such as the CBOE 10-Year Treasury Note Volatility Index (TYVIX) and the Treasury Implied Volatility (TIV) index constructed by Choi et al. (2017)—have been discontinued. This discontinuation was due to either limited market demand or, in the case of the TIV, the sophisticated nature of its construction and the inaccessibility of the requisite data. Nonetheless, we employ the TYVIX index as a robustness check and find our results to be consistent.

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covering daily DFG values from January 2004 to December 2022, we document a negatively predictive impact of DFG on subsequent stock market returns, which aligns with the implication of the model. To mitigate potential forward-looking biases arising from DFG measurements, we adopt alternate divergence measures and execute out-of-sample tests. Our results remain consistent across other robustness tests and at different intervals, including weekly and monthly predictions, both in-sample and out-of-sample.

Our study relates to the literature investigating the interrelation between bond and stock markets. One significant aspect of this literature delves into the "flight to safety" phenomenon, wherein, during periods of market stress, investors observe drastic, inverse movements in bond and equity markets. Such shifts result from investors reallocating their assets from the higher-risk equity market to the relatively safer Treasury bond market (Barsky 1989; Vayanos, 2004; Caballero and Krishnamurthy, 2008; Bekaert, Engstrom, and Xing 2009; Adrian, Crump, and Vogt, 2019; Baele, Bekaert, Inghelbrecht, and Wei, 2020). Notably, the DFG significantly differs from the "flight to safety" effect. In periods of "flight to safety," increased demand causes Treasury bond prices to rise, but concurrently, the MOVE index also surges in alignment with the VIX, suggesting heightened perceived market-wide volatility among Treasury bond investors. We demonstrate that the DFG exerts significant negative predictive effects, even when controlling for the "flight to safety" effect.

Numerous studies scrutinize the return co-movements (i.e., correlation) between bond and stock market returns (Baele, Bekaert, and Inghelbrecht, 2010; Yang, Zhou, Wang, 2010; Campbell, Pflueger, and Viceira, 2020; Li et al., 2022; Kozak, 2022). These investigations illuminate the linkage between bond and stock markets and their implications for investor behavior and asset pricing. A dynamic factor model proposed by Baele, Bekaert, and Inghelbrecht (2010) analyses the economic contributions of different factors to stock-bond return correlations and their time variations. Campbell, Pflueger, and Viceira (2020) introduced

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a new consumption-based asset pricing model with macroeconomic dynamics, aiming to understand the interplay between macroeconomics, stocks, and bonds, and their implications for bond-stock return co-movements. While these existing studies focus on documenting and explaining the dynamics of bond-stock return co-movements, there is rarely research on the predictive power of such co-movement. Our study extends this literature in two key ways. First, it examines the divergence of implied volatilities in bond and equity markets, thus presenting a novel measure to capture the expected volatility co-movements between bonds and stocks. This investigation transcends the traditional focus on bond-stock return co-movements and offers insightful findings about their volatility co-movements. Second, our paper pioneers in exploring the predictive power of DFG for stock market returns, as opposed to merely documenting its dynamics. This novel perspective illuminates the potential influence of DFG on future returns, providing meaningful insights for investors, policymakers, and practitioners.

The remainder of the paper is structured as follows: Section 2 presents the theoretical background and hypothesis. Section 3 outlines our data and methodology for empirical analysis. Section 4 discusses the empirical results, and Section 5 concludes.

2. Theoretical background and hypothesis

Bollerslev et al. (2009) develop a stylized self-contained general equilibrium model that incorporates the effects of time-varying economic uncertainty to demonstrate that the VRP, defined as the difference between model-free implied variance and realized variance, serves as a useful predictor of U.S. aggregate stock market returns. This predictive power arises because VRP captures the systematic risk associated with macroeconomic uncertainty and the volatility of that uncertainty (i.e., volatility-of-volatility) (Bollerslev et al. 2009; Drechsler & Yaron 2011; Drechsler 2013).

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As noted, both Treasury bond and equity markets respond to macroeconomic uncertainty, as reflected by the MOVE and VIX indices (e.g., Connolly, Stivers, and Sun, 2005; Baele, Bekaert, and Inghelbrecht, 2010; Hartmann, Straetmans, and de Vries, 2004; Greenwood, Hanson, and Stein, 2020). With different market characteristics, these two indices naturally present a time-varying divergence, which we refer to as the DFG. Using Bekaert and Hoerova's (2014) decomposition methodology, we decompose DFG into divergences in the conditional variance of stock returns (CV) and VRP of the Treasury and equity markets.[3] Consequently, DFG is expected to capture *common* macroeconomic uncertainty (i.e., the volatility of consumption growth) and the volatility-of-volatility, which also drive the equity risk premium (ERP), enabling DFG to predict future equity market returns.

We adopt and extend the equilibrium model from Bollerslev et al. (2009) to theoretically investigate the relationship between DFG and market returns. The full model development is presented in Appendix B, with key findings discussed in this section. In our model, we allow the consumption growth volatility to be stochastic, which leads to a two-factor structure for the aggregate stock market volatility and time-varying equity market VRP. According to model solutions given in Appendix B, the equity market VRP is a good equity market return predictor, as it is directly related to the volatility-of-volatility factor, q_t (see equation (A.11)), which appears in the ERP (see equation (A.19)) as highlighted by Bollerslev et al. (2009). The positive covariance between equity market VRP and ERP in Equation (A.22) suggests that the VRP positively predicts market returns.

Similarly, the DFG, as solved in Equation (A.18), contains the volatility-of-volatility factor, q_t, and should therefore predict equity market returns. DFG differs slightly from the

[3] Bekaert and Hoerova (2014) decompose the squared VIX index into the conditional variance of stock returns (CV) and the variance risk premium (VRP) to further investigate the VIX's predictive power. They find that while the VRP predicts stock returns, the CV forecasts economic activity and serves as a relatively better predictor of financial instability than the VRP.

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equity market VRP in Equation (A.11). In addition to q_t, DFG also captures the conditional variance of the growth rate $\sigma_{g,t}^2$, which appears in the ERP equation (A.19). This additional component in DFG further confirms that DFG could predict ERP and may imply different predictive power on market returns compared to VRP.[4] In this study, we empirically compare and discuss the return predictability of both VRP and DFG in Section 4.6.

The negative covariation between DFG and the ERP obtained in Equation (A.21) is primarily driven by $\theta < 0$; that is, the investor prefers the early resolution of uncertainty. θ is the key parameter in the recursive utility function that prices consumption uncertainty, resulting in the negative covariation between DFG and ERP. In addition, due to common macroeconomic uncertainties, the volatility of consumption growth and its volatility-of-volatility, DFG is expected to serve as a useful predictor for market returns. This negative covariation suggests that DFG has a significant negative predictive impact on equity market returns. Consequently, the following hypothesis is proposed:

Hypothesis*: The DFG negatively predicts future equity market returns.*

With the theoretical background provided by the equilibrium model in Appendix B, this paper primarily focuses on the empirical analysis of the DFG's predictive effects on future equity market returns.

3. Data

Our primary sample spans from January 1, 2004, to December 31, 2022, comprising 4,794 daily observations.[5] In the following subsections, we discuss our approach to quantifying

[4] It is worth mentioning that our study focuses on empirically testing the predictive effects of DFG and does not attempt to address how and to what extent the inclusion of the conditional variance of the growth rate could influence the market return predictability. This could be an interesting topic for future studies.

[5] Options on Treasury bonds and market indices were historically traded in open outcry pits, where the environment was often chaotic and noisy. This made it challenging for traders to obtain timely and accurate information about market conditions. On September 22, 2003, the CBOE Hybrid Trading System was

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the divergence between the fear gauges of the bond and stock markets, alongside other data pertinent to our analyses.

3.1. Divergence of fear gauges

We obtain daily data on MOVE and VIX from Bloomberg. Introduced in 1988, MOVE is a weighted average of one-month implied volatility on over-the-counter (OTC) traded Treasury options, weighted over two-, five-, ten-, and 30-year contracts for the subsequent 30 days. VIX represents the risk-neutral expected volatility of the stock market over the upcoming 30 days and is deduced from options on the S&P 500 index. Both indices depict investors' fear about future Treasury bond and equity markets. In our empirical analyses, we normalize these indices by dividing them by 100.

Due to the distinct original scaling of MOVE and VIX, to measure the divergence of the two fear gauges, we regress MOVE on VIX,

$$MOVE_t = \beta_0 + \beta_1 VIX_t + \varepsilon_t. \tag{1}$$

We utilize the residuals from Equation (1), $\widehat{\varepsilon_t}$, as our primary measure of the DFG. A positive divergence suggests that Treasury bond market investors anticipate higher volatility than stock market investors. Conversely, a negative DFG signifies a VIX level relatively higher than MOVE.

launched, which marked a significant step towards modernizing the trading process. This system combined traditional open outcry trading with electronic trading, laying the foundation for electronic trading of options. To mitigate any concerns that our results might be influenced by systematic liquidity improvements triggered by the introduction of electronic trading platforms, which might affect the pricing of fear gauges, particularly the VIX, we begin our sample period in January 2004. This starting date also allows for a buffer period during which investors can become more familiar with the new trading platform after its implementation.

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To address concerns that the residuals from regressing MOVE on VIX capture merely the idiosyncratic component of MOVE rather than divergence, we consider an alternative measure— the ratio of MOVE to VIX:

$$DFG_t^{ratio} = \frac{MOVE_t}{VIX_t}. \tag{2}$$

Despite these measures capturing DFG, it is worth noting that they do not completely align with the theoretical model's definition of DFG, where DFG is directly calculated as the difference between interest rate implied volatility and equity implied volatility. The primary reason is that MOVE measures the weighted average one-month implied volatility of options underlying various U.S. Treasury bonds across different maturities (two-year, five-year, ten-year, and 30-year) and is expressed in annualized basis points. This significantly differs from the calculation process of VIX, focusing solely on the implied volatility of S&P 500 index options. Consequently, simply dividing MOVE by 100 and then subtracting VIX may not provide any insightful information as the value will be overwhelmingly dominated by VIX.

To robustly test the implications of the model as well as the predictive effects of DFG, we consider an alternative interest rate volatility index, namely the CBOE 10-Year Treasury Note Volatility Index (TYVIX). TYVIX measures the market's expectation of 30-day interest rate volatility specifically focusing on the 10-year U.S. Treasury bond futures options and is expressed in annualized percentages, ensuring that the units and underlying mechanics of how volatility is estimated are comparable with those of VIX. We collect the data on TYVIX covering the period from January 2004 to December 2019 from.[6] Our analysis indicates that

[6] The data on TYVIX is available until May 2020. To mitigate concerns that the sample ending during the Covid-19 pandemic might distort the estimation, we end the test sample for TYVIX at the end of 2019. For robustness, we also test using TYVIX data up to May 2020 and document consistent results.

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the correlation coefficient between TYVIX and MOVE is 0.95. Using TYVIX, we develop a second alternative measure for DFG as the difference between TYVIX and VIX (*TYVIX_VIX*):

$$TYVIX_VIX_t = TYVIX_t - VIX_t. \quad (3)$$

Such a measure allows us to explicitly test the model implication.[7]

3.2. Other data

We employ S&P 500 index returns as a proxy for stock market returns, accessible from Bloomberg. In line with Da, Engelberg, and Gao (2015), we also incorporate the Economic Policy Uncertainty (*EPU*) index from Baker, Bloom and Davis (2016) and the Aruoba-Diebold-Scotti (*ADS*) business conditions index from the Federal Reserve Bank of Philadelphia as control variables. Data for the flight-to-safety dummy variable are obtained from Baele et al. (2020). We calculate the one-year rolling correlation between daily stock market returns and daily Treasury bond market return (*COR*) to assess stock-bond return co-movement. The daily returns of the Treasury bond market are represented by the Bloomberg Barclays U.S. Treasury Index. All variables are winsorized at the 1% level (0.5% in each tail) to mitigate outlier impacts.

Table 1 illustrates the summary statistics and correlation matrix for the pivotal variables. Figure 1 illustrates the co-movement between the two fear gauges (i.e., MOVE and VIX) throughout the sample period spanning January 2004 to December 2022.

TABLE 1 HERE

[7] Statistically, *TYVIX_VIX* is analogous to the residuals obtained from regressing TYVIX on VIX. This analogy holds true in terms of coefficient estimation and statistical significance when VIX is included as a control variable in predictive regressions. Similarly, this holds when comparing the regression results using our main DFG measure to those using a measure simply calculated as MOVE minus VIX, with VIX controlled.

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FIGURE 1 HERE

4. Empirical results

4.1. DFG and asset returns

To investigate the predictive effects of DFG at a daily frequency, we adopt the methodology from Da et al. (2015) and proceed with the following regressions:

$$Ret_{t+k} = \beta_0 + \beta_1 DFG_t + \sum_i \gamma_i Control_{i,t} + \varepsilon_{t+k}, \quad (4)$$

where Ret_{t+k} denotes the average cumulative daily stock market return over next k days, represented by the Standard and Poor's 500 index. Following Da et al. (2015), we utilize the average cumulative returns over subsequent two trading days, $Ret[t+1,t+2]$ as our primary metric. Furthermore, we extend our analysis to include cumulative daily returns over longer forecast horizons of five and ten days. Our control variables comprise lagged market returns (up to five lags), changes in the news-driven measure of economic policy uncertainty (*EPU*), the CBOE Volatility Index (*VIX*), and changes in the Aruoba-Diebold-Scotti (*ADS*) business conditions index. For all our findings, we calculate Newey-West standard errors and report the corresponding *t*-statistics. Additionally, for specifications with overlapped observations, we have examined Hodrick (1992) t-statistics and observed consistent results.[8]

TABLE 2 HERE

Table 2 reports the test results for the market return. We present the outcomes for univariate tests in columns (1), (3), (5), and (7), and those for multivariate regressions in columns (2), (4), (6), and (8). The negative coefficient on *DFG* indicates that higher DFG forecasts lower returns. Such predictive effects are statistically significant across all columns

[8] Results utilizing Hodrick (1992) *t*-statistics are available upon request.

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(k=1, 2, 5 and 10). Particularly, results from column (4) indicates that a standard-deviation surge in DFG forecasts a decrease of 5.2 basis points in the average daily market return over next two trading days (significant at the 1% level).

The baseline univariate regression exhibits an R^2 of 0.53% when we regress DFG on cumulative market returns over the subsequent two trading days. Given that the predictive regressions span a long period of 19 years with high-frequency daily data and that daily returns inherently possess a substantial unpredictable component, relatively modest R^2 statistics are anticipated. It is worth noting that the R^2 values from univariate predictive regressions are comparable to those from other well-known daily predictors, such as the Financial and Economic Attitude Revealed by Search (FEARS) index developed by Da et al. (2015). The univariate predictive regression of FEARS on stock market returns over the next two trading days presents an R^2 of 0.67%, but over a shorter sample period of eight years (from 2004 to 2011).

One potential concern might be that the predictive effect is exclusively driven by a single DFG component (i.e., MOVE or VIX). To alleviate this concern, we replace *DFG* with a simple component, either MOVE or VIX, in the predictive regressions together with other controls. We find no evidence suggesting that MOVE or VIX, considered as a singular component of DFG, exhibits statistically significant predictive power over the stock market returns. Specifically, untabulated results show that the coefficient of MOVE (VIX) for the average cumulative market return over next two trading days is -0.11 (0.12) with a t-statistics of -1.63 (0.42). These results confirm that the observed return predictability is indeed an output of the DFG, construed as an integrated function of MOVE and VIX.

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4.2. Robustness tests

In this section we test the robustness of our results against the incorporation of supplementary control variables and within alternative sample periods.

Some might argue that our main results are driven by the extreme daily stock market returns still present in the sample. For instance, in March 2020, we witnessed four circuit breakers triggered by the Covid-19 pandemic. These circuit breakers were also followed by substantial daily rallies. To address this concern, we perform two robustness tests. First, we apply a dummy for March 2020 to control for the considerable returns in that month. Second, we cut off our sample at the end of 2019, prior to the outbreak of COVID-19. The results for these two tests are reported in Panel A and B in Table 3, respectively. As illustrated in the table, the effects of DFG on future stock market returns, particularly the average market returns over subsequent two trading days, remain consistently significant.

TABLE 3 HERE

As an additional robustness check, we extend our sample back to 1990 since when the data on VIX is available. Even though this extended sample includes the period before and after the launch of the CBOE Hybrid Trading System, which could potentially influence systematic liquidity in the options market,we still find a consistent predictive effect of DFG and stock market returns, as presented in Panel C, Table 3. Moreover, since we have employed winsorization (0.5% in each tail) to mitigate the effects of outliers, a possible concern about applying winsorization in the context of predictive regressions is that it could introduce a forward-looking bias. To address this concern caused by winsorization, Panel D of Table 5 illustrates the results without winsorization. These results are again very similar to those

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reported in the earlier tables. Additionally, we control for the day-of-the-week effects in Panel E of Table 5, further corroborating the robustness of our results.

To comprehend the economic implications of DFG on future stock market returns, we delve into the return predictability of DFG on returns of tradable index securities. In alignment with Da et al. (2015), we employ four ETFs, namely the SPDR S&P 500 (NYSEARCA: SPY), the PowerShares QQQ Trust (NASDAQ: QQQ), the Russell 1000 Index ETF (NYSEARCA: IWB), and the Russell 2000 Index ETF (NYSEARCA: IWM). The findings presented in Panel F confirm significant predictive impact of DFG on returns over subsequent two trading days across all four ETFs, with regression coefficients resembling those reported in Table 2.

4.3. DFG and cross-sectional returns

In addition to stock market returns, we consider portfolios sorted by various factors and analyze whether the return predictive effect is also presented cross-sectionally. Following Da et al. (2015), we scrutinize portfolios sorted by beta and volatility. We source the data for the returns of beta-sorted and volatility-sorted portfolios from CRSP. We also include Fama-French size, BM and momentum portfolios in our tests to determine if the predictive effects of DFG align with those reported in earlier tables.[9]

TABLE 4 HERE

Table 4 documents the return predictability of univariate-sorted portfolios. Overall, the predictive effects of DFG remain consistent. Specifically, we find that DFG exerts a stronger predictive effect on returns among stocks with higher beta, higher volatility, smaller size,

[9] We obtain the data for Fama-French size, BM, and momentum portfolios from French's website.

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higher growth potential (low BM ratio), and those that experienced lower prior returns (lower momentum).

4.4. Alternative measures

Given that our primary measure for the divergence between MOVE and VIX is the residual of regressing MOVE on VIX, it could be argued that such a proxy may only reflect the idiosyncratic component of MOVE that is orthogonal to VIX, as opposed to the real divergence between the fear gauges. This concern should be partially allayed by the results reported in the earlier tables. To be explicit, if a Treasury bond market-specific volatility shock emerges, we would anticipate an increase in MOVE as Treasury bond investors expect higher volatility. However, such a bond-market-specific shock should not influence the stock market's movement or the VIX, and consequently would lead to a higher DFG. In other words, if DFG only captures bond-market-specific volatility, it should not exhibit any significant correlation with future stock market returns. However, our findings suggest that DFG is negatively related to future stock market returns, implying that DFG effectively measures the divergence rather than simply the idiosyncratic volatility of the Treasury bond market.

To further assuage this concern, we propose alternative measures for DFG. The first alternative measure of divergence is the ratio of MOVE to VIX (DFG^{ratio}). A higher DFG^{ratio} indicates that MOVE is greater relative to VIX. The test results using DFG^{ratio}, reported in Panel A of Table 5, align with those reported in Table 2. Another proxy for DFG we consider is *TYVIX_VIX*, calculated as the TYVIX minus VIX. Using an alternative interest rate volatility index allows us to further confirm the model's implications and the predictive effects of DFG. The results in Panel B of Table 5 indicate that a high *TYVIX_VIX*, which indicates that expected interest rate volatility (TYVIX) is relatively greater than expected equity market volatility (VIX), is associated with negative future stock market returns. Overall, these findings from

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alternative measures (i.e., DFG^{ratio} and *TYVIX_VIX*) are in line with the model and results reported in previous tables.

TABLE 5 HERE

4.5. DFG with flight-to-safety and stock-bond market co-movement

In this subsection, we compare DFG with other metrics for flight-to-safety and stock-bond market co-movement, which also reflect the interrelation between bond and stock markets. As a measure for flight-to-safety, we employ the FTS dummy from Baele et al. (2020). Furthermore, we use the one-year rolling correlation between daily bond and stock market returns as the co-movement indicator.

TABLE 6 HERE

As noted in Table 6, the FTS dummy and one-year rolling correlation between Treasury bond and stock market returns show no statistically significant correlation with stock market returns over subsequent trading days. In contrast, *DFG* still exhibits a significantly negative predictive impact on equity market returns. This suggests the predictive effects of DFG on stock market returns are not influenced by any other measures capturing the bond-stock market link, such as FTS and return correlation.

We then relax the FTS definition to define an alternative dummy, $FTS^{alternative}$, which equals one if the daily stock market return is less than -1%, while the daily Treasury bond market return is greater than zero. Additionally, we use one-month rolling correlation between daily Treasury bond and stock market returns (COR^{month}) as an alternative market co-movement measure. To further probe potential predictive effects of incorporating second moments of bond and stock market returns in the co-movement measure, we apply one-year and one-month

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rolling covariance between daily bond and stock market returns. Test results, reported in Appendix C, reconfirm the robustness of DFG's correlation with the future stock market returns.

4.6. DFG and risk aversion

As implied by the model, the predictive impact of DFG partially sources from the volatility-of-volatility factor which appears in the ERP as well as VRP. This raises the concern that the predictive power of DFG could be entirely subsumed by that of VRP. Moreover, fear gauges in some cases could be viewed as indicators of investor risk appetite, and VRP is widely recognized as a proxy for risk aversion, given that the constant relative risk aversion coefficient is proportionate to VRP (Bollerslev, Gibson, and Zhou, 2011). Consequently, a concern arises as to whether the predictability of DFG is solely derived from risk aversion. To address this issue, we incorporate VRP and an alternative proxy for risk aversion into our analyses, examining the relationship between DFG and future stock market returns. In line with Bollerslev, Tauchen, and Zhou (2009), we use five-minute S&P 500 index time series obtained from TICKDATA to calculate the daily realized variance. When calculating the conditional expectation of one-month variance, we apply a daily logarithmic HAR-RV-VIX model from Bekaert and Hoerova (2014) and Hu, Jacobs, and Seo (2022). Then VRP is calculated the difference between the ex ante physical variance estimates and the squared VIX. To measure risk aversion, we consider the daily Risk Aversion Index (*RA*) constructed by Bekaert, Engstrom and Xu (BEX, 2022), which is defined as the coefficient of relative risk aversion implied by the utility function and retrieved from the model and data on bonds and equities.

TABLE 8 HERE

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Table 8 presents our findings regarding the correlation between risk aversion and stock market returns. Panel A and B employ the *VRP* and the BEX *RA* index as the proxy for risk aversion, respectively. Our findings indicate that, in contrast to DFG, *VRP* and *RA* do not hold predictive capability for future stock market returns, thus alleviating concerns that the predictability of DFG observed was merely a manifestation of VRP or investors' risk aversion. In summary, the consideration of VRP and risk aversion reaffirms the efficacy of DFG as a predictor, and strengthens our conclusions drawn from equity market returns analysis.

4.7. Out-of-sample test

Concerns may araise regarding the potential forward-looking biases involved in *DFG* which are obtained from the whole-sample regression. To alleviate this concern as well as those about overfitting and finite sample biases, we conduct out-of-sample tests for *DFG* and other factors discussed in our study. To this end, we adopt an expanding window following Liu and Matthies (2022) for coefficient estimation, which allows the use of more available data. To ensure a sizable quantity of daily observations for our initial in-sample training and to secure a substantial out-of-sample period for forecast evaluation, we launch our coefficient estimation for the out-of-sample forecast with a minimum of 2,016 trading days (eight years) of data.

Drawing from Welch and Goyal (2008), our out-of-sample forecasts at time *t* utilize only information available up until *t* to recursively estimate the "out-of-sample" DFG and predict stock market returns over the forthcoming *k* periods. Specifically, to forecast the return over next k periods at time *t*, we begin by regressing MOVE on VIX to yield an estimated *DFG* using data up until time *t-k* ($\widehat{DFG_{t-k,t}}$). We then regress ex post future stock market returns on the lagged estimated *DFG*, which are solely available up to time *t-k*, to determine the

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coefficient of the lagged estimated *DFG* ($\mu_{t-k,t}$). We subsequently conduct another regression of MOVE on VIX using data up until time t to obtain the most recent *DFG* estimation ($\widehat{DFG}_{t,t}$). Using $\mu_{t-k,t}$ based on period [0, *t*-*k*] and the latest $DFG_{t,t}$ based period [0, *t*], we forecast the future stock market returns over the time interval [*t*+1, *t*+*k*] ($\hat{r}_{t+k}$). We calculate the out-of-sample R^2 as:

$$R_{OOS}^2 = 1 - \frac{\sum_{t=p}^{T-k}\left(r_{t+k} - \hat{r}_{t+k}\right)^2}{\sum_{t=p}^{T-k}\left(r_{t+k} - \bar{r}_{t+k}\right)^2}, \tag{5}$$

where $\hat{r}_{t+k}$ is the forecasted *k*-period-ahead cumulative stock market return, which is estimated using information up to time *t*. $\bar{r}_{t+k}$ symbolizes the historical mean of *k*- period-ahead cumulative stock market return up to time *t,* and *T* represents the sample size. If a variable proves to be a robust predictor, we expect to secure an out-of-sample R^2 that is statistically greater than zero. To further validate the statistical significance of the out-of-sample R^2, we apply the mean squared forecast error (MSFE)-adjusted statistic of Clark and West (2007).[10] Consistent with our im-sample test, we focus on return forecast over subsequent two trading days.

TABLE 9 HERE

Table 9 exhibits the results of the univariate out-of-sample analysis. It is evident that *DFG* is the predictor that achieves the highest positive out-of-sample R^2 of 0.26%. This outcome aligns well its in-sample R^2 of 0.23%, confirming DFG as a potent predictor of stock market returns. The MSFE-adjusted t-statistic of 2.5 implies that the out-of-sample daily

[10] Accordingly, the MSFE-adjusted statistic tests the null hypothesis that the historical average MSFE does not surpass the predictive regression forecast MSFE against the one-sided alternative hypothesis that the historical average MSFE exceeds the predictive regression forecast MSFE.

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predictive effect of DFG holds statistical significance. These findings underscore that the out-of-sample predictive significance of DFG surpasses that of other factors. Furthermore, *VIX*, *Ret*, *FTS*, *COR*, *VRP* and *RA* fail to exhibit any significant out-of-sample predictive capacity. Although the out-of-sample R^2 of MOVE (0.14%) is deemed significantly positive only at a 10% significance level, with an MSFE t-statistics of 1.77, such an outcome contradicts its in-sample forecast result discussed in Section 4.1., and thus, should be approached with prudence.

4.8. Weekly and monthly forecast

In this section, we extend our investigation by examining weekly and monthly data for DFG estimation and return forecast analysis. This approach helps filter out the noise from daily fluctuations, allowing us to focus on significant volatility patterns prevalent in the broader markets.

TABLE 10 HERE

In Table 10, Panel A and B show the forecast tests on a weekly and monthly basis, respectively. Consistent with our daily DFG and stock market return analysis, we conduct both in-sample and out-of-sample tests for forecast horizons extending to 24 periods. For the out-of-sample tests, we maintain an initial eight-year training period (416 weeks/96 months). The results suggest that DFG negatively predicts stock market returns on both a weekly and monthly basis in-sample. The out-of-sample R^2s are all positive with statistically significant MSFE *t*-statistics, which corroborate in-sample findings. As anticipated, both in-sample and out-of-sample R^2 values are significantly higher when using data of a lower frequency. These outcomes, drawn from weekly and monthly data, further substantiate our claim that DFG negatively predicts equity market returns.

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For robustness, we follow previous studies (e.g., Bollerslev, Tauchen, and Zhou, 2009;) and include economic variables in our monthly predictive regressions. Specifically, we employ 14 economic variables and their first principal component to compare the forecasting power of DFG with these economic predictors. This approach allows us to examine whether the forecasting power of DFG is driven by omitted economic variables related to business cycles or changes in macroeconomic risks. The 14 monthly economic variables include the following: earnings-price ratio, *EP*, defined as the difference between the log of earnings on the S&P 500 index and the log of prices; dividend-payout ratio, *DE*, defined as the difference between the log of dividends and the log of earnings on the S&P 500 index; dividend-price ratio, *DP*, defined as the difference between the log of a 12-month moving sum of dividends paid on the S&P 500 index and the log of S&P 500 index price; dividend yield, *DY*, defined as the difference between the log of S&P 500 dividends and the log of lagged S&P 500 prices; stock return variance, *SVAR*, calculated as the sum of squared daily returns on the S&P 500 index; book-to-market ratio, *BM*, defined as the ratio of book value to market value for the Dow Jones Industrial Average; net equity expansion, *NTIS*, calculated as the 12-month moving sums of net issues by stocks listed on NYSE divided by the total end-of-year market capitalization of NYSE stocks; Treasury bill rate, *TBL*, defined as the yield of a 3-month T-bill; long-term yield, *LTY*, which is the long-term government bond yield; long-term return, *LTR*, defined as the return on long-term government bonds; term spread, *TMS*, calculated as the long-term yield minus the T-bill rate; default yield spread, *DFY*, defined as the difference between BAA- and AAA-rated bond yields; default return spread, *DFR*, calculated as the difference between the long-term corporate bond return and the long-term government bond return; inflation, *INFL*, calculated based on the Consumer Price Index by Welch and Goyal (2008). The data required

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for calculation of all 14 economic variables above are sourced from Goyal's website.[11] Additionally, we consider the first principal component of these 14 variables, *ECON*.

We first consider the predictive regression on a single economic variable and report the results in Panel A, Table A.2 in Appendix D. Out of the 15 individual economic predictors, only book-to-market ratio (*BM*) and long-term yield (*LTY*) exhibit significant predictive abilities for the market returns at 10% significance levels. These two significant economic variables present R^2s of 1.5% and 1.2%, respectively. We then investigate whether the forecasting power of DFG remains significant after controlling for economic predictors. To analyze the incremental forecasting power of DFG, we conduct bivariate predictive regressions that include both *DFG* and each economic variable, with the results presented in Panel B of Table A.2. The results show that the coefficient estimates of *DFG* remain statistically significant even when augmented by the economic predictors. The R^2 values range from 5.9% to 8.8%, substantially larger than those reported in Panel A, which are based solely on the economic predictors. Thus, *DFG* outperforms all 14 individual economic predictors and their first principal component (*ECON*) in forecasting the monthly market returns in-sample. These findings demonstrate that the return predictability of DFG is not driven by macroeconomic fundamentals, further affirming DFG's predictive effects on equity market returns.

Next, we compare DFG with other predictors discussed in previous sections in monthly market return forecast and report the results in Appendix E. The findings suggest DFG's predictive power remains significant when other monthly predictors are included. Additionally, to address concerns about the relatively short sample period for the monthly forecast, we

[11] https://sites.google.com/view/agoyal145

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conduct an extended analysis covering the period from 1990 to 2022. The results, presented in Appendix F, provide consistent evidence confirming the predictive effects of DFG.

5. Conclusion

Motivated by heightened attention to interest rate volatility and arguments from practitioners as reported by well-known financial media, this study explores the relationship between the divergence of fear gauges, the MOVE and VIX indices, and stock market returns. In an equilibrium model, we demonstrate that there is a negative relationship between the DFG and future market returns. Consistent with the model's implications, our empirical analysis reveals that DFG has a significant predictive effect on future aggregate stock market returns and the cross-section of stock returns. Additionally, DFG's predictive power surpasses other variables that capture the link between Treasury bond and equity markets, such as the measures for flight to safety and correlation of market returns. Furthermore, it outperforms measures of risk aversion and commonly used macroeconomic variables.

Our study contributes to the literature on the connection and co-movements between bond and stock markets by introducing a novel perspective on the role of fear gauges and their divergence. Additionally, since DFG is a readily available market-based measure at a high frequency, we offer valuable implications for various stakeholders in their decision-making processes.

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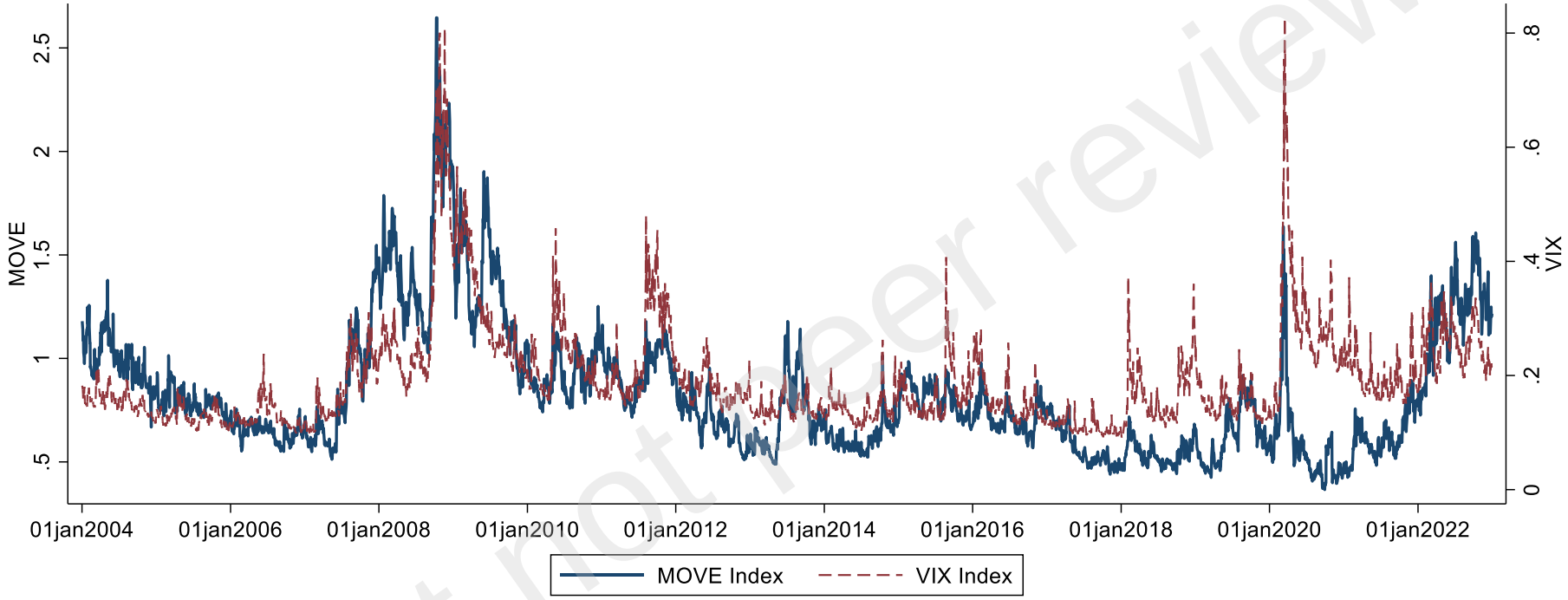
Fig. 1



This figure presents the co-movement of the MOVE and VIX indices over the sample period January 2004 to December 2022.

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Table 1 Summary Statistics

Panel A: Summary statistics

	Obs.	Mean	SD	Min	Max	p50	P25	P75
Ret (%)	4794	0.0338	1.0904	-4.0291	3.9059	0.0679	-0.4132	0.5603
DFG	4794	-0.0082	0.2357	-1.0338	0.9973	-0.0293	-0.1375	0.1148
DFG^{ratio}	4794	0.0459	0.0129	0.0157	0.0843	0.0461	0.0371	0.0549
TYVIX_VIX	4111	-0.1253	0.0783	-0.7224	-0.0476	-0.1005	-0.1433	-0.0787
MOVE	4794	0.8279	0.3117	0.3660	2.6460	0.7510	0.6060	0.9580
VIX	4794	0.1927	0.0894	0.0914	0.8269	0.1662	0.1335	0.2250
EPU	4794	-0.0673	0.5855	-3.8167	3.9367	-0.0496	-0.3452	0.2254
ADS	4794	0.0000	0.1415	-3.3680	3.6166	-0.0018	-0.0123	0.0111
FTS	4794	0.0240	0.1530	0.0000	1.0000	0.0000	0.0000	0.0000
COR	4794	-0.3038	0.1994	-0.6789	0.1660	-0.3461	-0.4560	-0.1354
VRP	4794	1.1100	1.5853	-5.7348	21.3171	0.6417	0.3912	1.2254
RA	4794	3.1244	1.4882	2.4254	32.7108	2.7986	2.6284	3.1361

Panel B: Correlation matrix

	DFG	DFG^{ratio}	TYVIX_VIX	MOVE	VIX	EPU	ADS	FTS	COR	VRP
DFG^{ratio}	0.723									
TYVIX_VIX	0.093	0.597								
MOVE	0.728	0.164	-0.613							
VIX	0.056	-0.486	-0.986	0.726						
EPU	0.011	0.011	0.006	0.005	-0.004					
ADS	-0.017	-0.029	0.090	-0.079	-0.098	-0.007				
FTS	0.066	-0.080	-0.181	0.175	0.188	0.000	-0.074			
COR	-0.016	0.321	0.347	-0.257	-0.358	0.028	0.009	-0.068		
VRP	0.028	-0.390	-0.908	0.648	0.915	-0.011	-0.085	0.105	-0.267	
RA	0.033	-0.298	-0.801	0.577	0.807	-0.012	-0.174	0.092	-0.229	0.873

This table presents the summary statistics (Panel A) and the correlation matrix (Panel B) for the principal variables utilized in our research. A comprehensive description of these variables is provided in Appendix A.

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Table 2 Baseline Regression Results

	(1) Ret(t+1)	(2) Ret(t+1)	(3) Ret[t+1, t+2]	(4) Ret[t+1, t+2]	(5) Ret[t+1, t+5]	(6) Ret[t+1, t+5]	(7) Ret[t+1, t+10]	(8) Ret[t+1, t+10]
DFG	-0.241***	-0.216**	-0.236***	-0.223***	-0.225***	-0.215***	-0.236***	-0.233***
	(-2.84)	(-2.45)	(-3.25)	(-3.00)	(-3.87)	(-3.58)	(-4.28)	(-4.15)
VIX		0.163		0.130		0.122		0.180
		(0.48)		(0.47)		(0.56)		(0.94)
Ret(t)		-0.081***		-0.042***		-0.026***		-0.009
		(-3.75)		(-2.71)		(-3.01)		(-1.59)
Ret(t-1)		-0.009		-0.007		-0.022**		-0.004
		(-0.36)		(-0.44)		(-2.54)		(-0.65)
Ret(t-2)		0.326		-1.528		-1.163		-0.191
		(0.13)		(-0.98)		(-1.36)		(-0.32)
Ret(t-3)		-1.883		-2.284		-1.598**		-0.276
		(-0.83)		(-1.50)		(-2.00)		(-0.47)
Ret(t-4)		-2.538		-3.062**		-0.776		-0.119
		(-1.09)		(-2.10)		(-0.94)		(-0.21)
EPU		-0.082**		-0.010		-0.000		-0.005
		(-2.49)		(-0.50)		(-0.02)		(-0.85)
ADS		-0.082		0.131		0.182		0.071
		(-0.43)		(0.86)		(1.36)		(0.70)
_cons	0.032**	-0.001	0.032**	0.010	0.032***	0.011	0.031***	-0.003
	(2.11)	(-0.01)	(2.45)	(0.21)	(2.80)	(0.30)	(2.85)	(-0.09)
Adj.R^2	0.0024	0.0111	0.0053	0.0117	0.0137	0.0244	0.0324	0.0366
N	4794	4794	4794	4794	4794	4794	4794	4794

This table illustrates the relationship between daily cumulative stock market returns and DFG. *DFG* measures the divergence of fear gauges and is calculated as the residuals of regressing MOVE on VIX. Control variables include lagged returns up to five lags, changes in a news-based measure of economic policy uncertainty (EPU), the CBOE volatility index (VIX), and changes in the Aruoba-Diebold-Scotti (ADS) business conditions index. Newey-West standard errors,

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with the lag number equal to the forecast horizons, are estimated, and the corresponding t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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Table 3 Robustness Tests

Panel A: With dummy for COVID-19

	(1) Ret(t+1)	(2) Ret[t+1, t+2]	(3) Ret[t+1, t+5]	(4) Ret[t+1, t+10]
DFG	-0.283***	-0.273***	-0.247***	-0.244***
	(-3.29)	(-3.81)	(-4.16)	(-4.41)
Controls	YES	YES	YES	YES
Adj.R^2	0.0171	0.0191	0.0330	0.0386
N	4794	4794	4794	4794

Panel B: Subsample 2004-2019

	(1) Ret(t+1)	(2) Ret[t+1, t+2]	(3) Ret[t+1, t+5]	(4) Ret[t+1, t+10]
DFG	-0.201*	-0.200**	-0.173**	-0.218***
	(-1.77)	(-2.14)	(-2.35)	(-3.17)
Controls	YES	YES	YES	YES
Adj.R^2	0.0109	0.0118	0.0192	0.0284
N	4032	4032	4032	4032

Panel C: Extended sample 1990-2022

	(1) Ret(t+1)	(2) Ret[t+1, t+2]	(3) Ret[t+1, t+5]	(4) Ret[t+1, t+10]
DFG	-0.095*	-0.103**	-0.098***	-0.105***
	(-1.73)	(-2.20)	(-2.85)	(-4.26)
Controls	YES	YES	YES	YES
Adj.R^2	0.0034	0.0068	0.0150	0.0125
N	8347	8347	8347	8347

Panel D: Without winsorization

	(1) Ret(t+1)	(2) Ret[t+1, t+2]	(3) Ret[t+1, t+5]	(4) Ret[t+1, t+10]
DFG	-0.253**	-0.239***	-0.240***	-0.256***
	(-2.36)	(-2.78)	(-3.91)	(-6.20)
Controls	YES	YES	YES	YES
Adj.R^2	0.0226	0.0164	0.0260	0.0385
N	4794	4794	4794	4794

Panel E: Control for the day of the week effects

	(1) Ret(t+1)	(2) Ret[t+1, t+2]	(3) Ret[t+1, t+5]	(4) Ret[t+1, t+10]
DFG	-0.216**	-0.222***	-0.215***	-0.233***
	(-2.44)	(-3.00)	(-4.14)	(-6.46)
Controls	YES	YES	YES	YES
Adj.R^2	0.0108	0.0116	0.0236	0.0358
N	4794	4794	4794	4794

Panel F: Tradability with ETFs

	(1) SPY	(2) Russell1000	(3) Russell2000	(4) QQQ
DFG	-0.237***	-0.240***	-0.250**	-0.273***
	(-2.73)	(-2.81)	(-2.36)	(-2.96)
Controls	YES	YES	YES	YES
Adj.R^2	0.0133	0.0126	0.0102	0.0106
N	4794	4794	4794	4794

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This table reports the results from various robustness checks. In Panel A-E, the dependent variables are future daily cumulative equity market returns. Panel A provides a robustness check with the inclusion of a COVID-specific dummy variable for March 2020. Panel B examines a sub-sample period spanning from 2004 to 2019, thus excluding the COVID period. Panel C extends the sample to include data from January 1990, the commencement of VIX data availability, through December 2022. Panel D shows the results without winsorization. Panel E includes the day of the week effects in addition to the existing controls. Panel F considers tradability, presenting results based on the open-to-close adjusted returns over the next two trading days for various exchange traded funds. These ETFs include the S&P Exchange Traded Fund (SPY), the Russell 1000 Exchange Traded Fund (IWB), the Russell 2000 Exchange Traded Fund (IWM), and the NASDAQ Exchange Traded Fund (QQQ). *DFG* is the measure of divergence of fear gauges calculated as the residuals of regressing MOVE on VIX. All specification includes a set of controls, including lagged returns up to five lags, changes in a new-based measure of economic policy uncertainty (EPU), the CBOE volatility index (VIX), and changes in the Aruoba-Diebold-Scotti (ADS) business conditions index. Newey-West standard errors, with the lag number equal to the forecast horizons, are estimated, and the corresponding t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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Table 4 DFG and Cross-Sectional Returns

Panel A: Beta portfolio

	Low-beta	2	3	4	5	6	7	8	9	High-beta
DFG	-0.083**	-0.123***	-0.156**	-0.192**	-0.196**	-0.219**	-0.232**	-0.260**	-0.318**	-0.340**
	(-2.04)	(-2.70)	(-2.39)	(-2.51)	(-2.27)	(-2.33)	(-2.27)	(-2.28)	(-2.46)	(-2.15)
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.0624	0.0299	0.0105	0.0089	0.0083	0.0080	0.0088	0.0075	0.0084	0.0125

Panel B: Volatility portfolio

	Low-volatility	2	3	4	5	6	7	8	9	High-volatility
DFG	-0.082**	-0.138**	-0.151**	-0.164*	-0.205**	-0.227**	-0.256**	-0.292**	-0.333***	-0.295**
	(-2.50)	(-2.47)	(-2.14)	(-1.96)	(-2.15)	(-2.27)	(-2.37)	(-2.57)	(-2.72)	(-2.41)
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.0681	0.0141	0.0074	0.0064	0.0066	0.0065	0.0069	0.0100	0.0193	0.0377

Panel C: Size portfolio

	Small	2	3	4	5	6	7	8	9	Big
DFG	-0.264***	-0.323***	-0.262**	-0.242**	-0.249**	-0.214**	-0.236***	-0.243***	-0.213**	-0.218***
	(-2.98)	(-3.09)	(-2.51)	(-2.42)	(-2.54)	(-2.31)	(-2.62)	(-2.77)	(-2.57)	(-2.92)
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.0098	0.0080	0.0070	0.0074	0.0077	0.0073	0.0074	0.0085	0.0084	0.0132

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Panel D: BM portfolio

	Low	2	3	4	5	6	7	8	9	High
DFG	-0.269***	-0.214***	-0.205***	-0.195**	-0.196**	-0.239***	-0.189**	-0.209**	-0.163	-0.212*
	(-3.23)	(-2.84)	(-2.84)	(-2.44)	(-2.38)	(-2.78)	(-2.02)	(-2.04)	(-1.52)	(-1.69)
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.0096	0.0106	0.0122	0.0079	0.0098	0.0081	0.0056	0.0084	0.0069	0.0047

Panel E: Momentum portfolio

	Loser	2	3	4	5	6	7	8	9	Winner
DFG	-0.387**	-0.348***	-0.291***	-0.247**	-0.226**	-0.200**	-0.169**	-0.199***	-0.193**	-0.172*
	(-2.33)	(-2.69)	(-2.64)	(-2.49)	(-2.48)	(-2.39)	(-2.22)	(-2.70)	(-2.53)	(-1.75)
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.0149	0.0105	0.0072	0.0081	0.0098	0.0109	0.0087	0.0092	0.0094	0.0055

This table links DFG to the daily returns of portfolios constructed by sorting on stock characteristics related to limits to arbitrage. In Panel A, B, C, D, and E, portfolios are sorted by the CAPM beta, the total volatility, size, book to market ratio and momentum, respectively. The dependent variable is the cumulative portfolio returns over subsequent two trading days. *DFG* is the measure of divergence of fear gauges calculated as the residuals of regressing MOVE on VIX. The set of control variables (unreported) include lagged returns up to five lags, changes in a new-based measure of economic policy uncertainty (EPU), the CBOE volatility index (VIX), and changes in the Aruoba-Diebold-Scotti (ADS) business conditions index. Newey-West t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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Table 5 Alternative Divergence Measures

Panel A: Divergence based on the ratio of MOVE to VIX

	(1) Ret(t+1)	(2) Ret[t+1, t+2]	(3) Ret[t+1, t+5]	(4) Ret[t+1, t+10]
DFG^{ratio}	-2.932**	-3.371***	-3.469***	-3.639***
	(-1.98)	(-2.76)	(-3.33)	(-3.77)
Controls	YES	YES	YES	YES
Adj.R^2	0.0099	0.0095	0.0195	0.0230
N	4794	4794	4794	4794

Panel B: Divergence based on the difference between TYVIX and VIX

	(1) Ret(t+1)	(2) Ret[t+1, t+2]	(3) Ret[t+1, t+5]	(4) Ret[t+1, t+10]
TYVIX_VIX	-2.153	-2.361*	-1.878*	-2.984***
	(-1.36)	(-1.81)	(-1.70)	(-2.77)
Controls	YES	YES	YES	YES
Adj.R^2	0.0101	0.0104	0.0162	0.0238
N	4017	4017	4017	4017

This table presents the test results using alternative measures for the divergence of fear gauges. In Panel A, DFG^{ratio} is defined as the ratio of MOVE to VIX. In panel B, *TYVIX_VIX* is defined as the difference between TYVIX and VIX. The set of control variables (unreported) include lagged returns up to five lags, changes in a new-based measure of economic policy uncertainty (EPU), the CBOE volatility index (VIX), and changes in the Aruoba-Diebold-Scotti (ADS) business conditions index. Newey-West standard errors, with the lag number equal to the forecast horizons, are estimated, and the corresponding t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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Table 6 DFG With Flight-To-Safety and Stock-Bond Market Correlation

Pane A: FTS and DFG

	(1) Ret(t+1)	(2) Ret(t+1)	(3) Ret[t+1, t+2]	(4) Ret[t+1, t+2]	(5) Ret[t+1, t+5]	(6) Ret[t+1, t+5]	(7) Ret[t+1, t+10]	(8) Ret[t+1, t+10]
FTS	0.194	0.225	0.129	0.159	-0.028	-0.000	-0.041	-0.010
	(1.24)	(1.43)	(1.12)	(1.39)	(-0.36)	(-0.00)	(-0.61)	(-0.15)
DFG		-0.231***		-0.232***		-0.213***		-0.237***
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.0096	0.0119	0.0075	0.0127	0.0119	0.0243	0.0053	0.0385
N	4794	4794	4794	4794	4794	4794	4794	4794

Pane B: COR and DFG

	(1) Ret(t+1)	(2) Ret(t+1)	(3) Ret[t+1, t+2]	(4) Ret[t+1, t+2]	(5) Ret[t+1, t+5]	(6) Ret[t+1, t+5]	(7) Ret[t+1, t+10]	(8) Ret[t+1, t+10]
COR	-0.015	0.011	-0.016	0.011	-0.014	0.011	0.008	0.036
	(-0.20)	(0.15)	(-0.25)	(0.18)	(-0.26)	(0.21)	(0.15)	(0.70)
DFG		-0.221**		-0.225***		-0.214***		-0.240***
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.0089	0.0110	0.0069	0.0117	0.0118	0.0243	0.0049	0.0390
N	4794	4794	4794	4794	4794	4794	4794	4794

This table reports the test results considering the flight-to-safety dummy in Panel A and bond-stock market co-movement in Panel B, respectively. *FTS* is the dummy constructed by Baele et al. (2020). *COR* is the one-year rolling correlation between daily Treasury bond and stock market returns. *DFG* is the measure of divergence of fear gauges calculated as the residuals of regressing MOVE on VIX. The set of control variables (unreported) include lagged returns up to five lags, changes in a new-based measure of economic policy uncertainty (EPU), the CBOE volatility index (VIX), and changes in the Aruoba-Diebold-Scotti (ADS) business conditions index. Newey-West standard errors, with the lag number equal to the forecast horizons, are estimated, and the corresponding t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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Table 8 DFG and Risk Aversion

Panel A: Variance risk premium

	(1) Ret(t+1)	(2) Ret(t+1)	(3) Ret[t+1, t+2]	(4) Ret[t+1, t+2]	(5) Ret[t+1, t+5]	(6) Ret[t+1, t+5]	(7) Ret[t+1, t+10]	(8) Ret[t+1, t+10]
VRP	0.072	0.068	0.050	0.046	0.023	0.019	0.023	0.019
	(1.42)	(1.35)	(1.46)	(1.34)	(0.88)	(0.73)	(1.02)	(0.88)
DFG		-0.211**		-0.218***		-0.210***		-0.235***
		(-2.41)		(-2.93)		(-3.54)		(-4.24)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.0108	0.0127	0.0089	0.0134	0.0129	0.0251	0.0075	0.0403
N	4794	4794	4794	4794	4794	4794	4794	4794

Panel B: Risk aversion index by BEX

	(1) Ret(t+1)	(2) Ret(t+1)	(3) Ret[t+1, t+2]	(4) Ret[t+1, t+2]	(5) Ret[t+1, t+5]	(6) Ret[t+1, t+5]	(7) Ret[t+1, t+10]	(8) Ret[t+1, t+10]
RA	0.044	0.047	0.007	0.010	0.007	0.010	0.007	0.010
	(1.10)	(1.17)	(0.24)	(0.33)	(0.38)	(0.53)	(0.47)	(0.78)
DFG		-0.228**		-0.226***		-0.215***		-0.239***
		(-2.57)		(-3.02)		(-3.60)		(-4.29)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.0103	0.0125	0.0070	0.0119	0.0120	0.0247	0.0053	0.0393
N	4794	4794	4794	4794	4794	4794	4794	4794

This table reports the results for the predictive effects of DFA together with VRP and risk aversion on stock market returns. In Panel A, VRP in included and estimated as the difference between the physical and risk-neutral expectations of the future return variance. In Panel B, risk aversion is included and measured by the daily Risk Aversion Index (*RA*) constructed by Bekaert, Engstrom and Xu (2022), defined as the coefficient of relative risk aversion implied by the utility

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function and retrieved from the model and data on corporate bonds and equities. *DFG* is the measure of divergence of fear gauges calclated as the residuals of regressing MOVE on VIX. The set of control variables include lagged stock market returns up to five lags, changes in news–based measure of economic policy uncertainty (EPU), the CBOE volatility index (VIX), and changes in the Aruoba–Diebold–Scotti (ADS) business conditions index. Newey-West t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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Table 9 Out-of-sample Predictability

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	DFG	VIX	MOVE	Ret	FTS	COR	VRP	RA
OOS R^2	0.0026	0.0002	0.0014	-6E-04	-0.0010	-0.0011	-0.0002	-0.0045
MSFE t-stat	2.489	0.571	1.772	1.165	0.011	-2.451	0.327	-1.310

This table presents the out-of-sample test results for forecasting the cumulative stock market return over next two trading days. Out-of-sample R square and the mean squared forecast error (MSFE)-adjusted statistic of Clark and West (2007) are reported for univariate predictive regressions. *DFG* is the measure of divergence of fear gauges. *VIX* is the Chicago Board Options Exchange Volatility Index. *MOVE* is the Merrill Lynch Option Volatility Estimate index. *Ret* is the current daily stock market return. *FTS* is the dummy constructed by Baele et al. (2020). *COR* is the one-year rolling correlation between daily Treasury bond and stock market returns. VRP is estimated as the difference between the physical and risk-neutral expectations of the future return variance. *RA* is the daily Risk Aversion Index (*RA*) constructed by Bekaert, Engstrom and Xu (2022), defined as the coefficient of relative risk aversion implied by the utility function and retrieved from the model and data on corporate bonds and equities. The initial training period is eight years (2,016 trading days).

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Table 10 Weekly and Monthly Forecast

Pane A: Weekly forecast

	(1)	(2)	(3)	(4)	(5)	(6)
Forecast horizons	w=1	w=3	w=6	w=9	w=12	w=24
DFG	-1.282***	-1.229***	-0.988***	-0.920***	-0.870***	-0.811***
	(-2.99)	(-3.56)	(-3.47)	(-3.41)	(-3.43)	(-3.24)
_cons	0.001**	0.001**	0.001***	0.001***	0.001***	0.001***
	(2.00)	(2.36)	(2.60)	(2.74)	(2.80)	(2.82)
Adj. R^2	0.0141	0.0452	0.0643	0.0877	0.1099	0.1833
N	992	992	991	988	985	973
OOS R^2	0.0119	0.0330	0.0782	0.1352	0.1761	0.2978
MSFE t-stat	1.9688	3.4371	5.2825	6.2172	7.2498	8.4938

Pane B: Monthly forecast

	(1)	(2)	(3)	(4)	(5)	(6)
Forecast horizons	m=1	m=3	m=6	m=9	m=12	m=24
DFG	-4.634***	-3.738***	-3.409***	-3.008***	-2.717***	-0.953**
	(-2.86)	(-3.36)	(-3.34)	(-2.71)	(-2.75)	(-2.13)
_cons	0.006**	0.006**	0.006***	0.006***	0.006***	0.008***
	(2.19)	(2.59)	(2.74)	(2.82)	(2.86)	(3.57)
Adj. R^2	0.0630	0.1343	0.1984	0.2216	0.2383	0.0560
N	228	228	225	222	219	207
OOS R^2	0.0403	0.2160	0.2944	0.3043	0.3171	0.1442
MSFE t-stat	1.7868	3.6127	4.1413	3.9942	4.0290	3.1469

This table presents the test results for the predictive effects of DFG on weekly stock market returns (in Panel A) and monthly stock market returns (in Panel B) both in-sample and out-of-sample. *DFG,* the divergence of fear gauges, is quantified as the residuals derived from regressing MOVE on VIX, computed on a

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weekly or monthly basis. For in-sample results, Newey-West t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Out-of-sample R square and the mean squared forecast error (MSFE)-adjusted statistic of Clark and West (2007) are reported. The initial training period is eight years (416 weeks or 96 months).

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Appendix A: Variable definition

Variable	Description
DFG	Main measure of the divergence between the bond market fear gauge (MOVE) and stock market fear gauge (VIX), estimated as the residuals of regressing MOVE on VIX.
DFG^{ratio}	Alternative measure of the divergence of fear gauges, estimated as the ratio of MOVE to VIX.
TYVIX_VIX	Alternative measure of the divergence of fear gauges, calculated as the CBOE 10-Year Treasury Note Volatility Index (TYVIX) minus VIX
MOVE	Merrill Lynch Option Volatility Estimate index measuring the expected volatility of Treasury bond market.
VIX	Chicago Board Options Exchange Volatility Index
Ret	Stock mark return proxied by S&P 500 index
R_Treasury	Treasury bond market return proxied by Bloomberg Barclays U.S. Treasury Index
EPU	Changes in the new-based measure of economic policy uncertainty (EPU) index
ADS	Changes in the Aruoba-Diebold-Scotti (ADS) business conditions index
FTS	Flight-to-safety dummy developed by Baele, Bekaert, and Inghelbrecht (2010)
COR	Bond-stock market co-movement measure, calculated as the one-year rolling correlation between daily bond market returns and the daily stock market returns
RA	Daily risk aversion index developed by Bekaert, Engstrom and Xu (2022)
VRP	Variance risk premium calculated as difference between the physical and risk-neutral expectations of the future return variance
$FTS^{alternative}$	Alternative measure for flight-to-safety. It is a dummy equal to one if the daily stock market return is lower than -1% while the daily Treasury bond market return is greater than zero.
$CORet^{onth}$	Alternative measure for bond-stock market co-movement, calculated as one-month rolling correlation between daily bond and stock market returns
Covariance	Alternative measure for bond-stock market co-movement, calculated as one-year rolling covariance between daily bond and stock market returns
$Covariance^{month}$	Alternative measure for bond-stock market co-movement, calculated as one-month rolling covariance between daily bond and stock market returns
EP	Earnings-price ratio defined as the difference between the log of earnings on the S&P 500 index and the log of prices
DE	Dividend-payout ratio, defined as the difference between the log of dividends and the log of earnings on the S&P 500 index

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DP	Dividend-price ratio, defined as the difference between the log of a 12-month moving sum of dividends paid on the S&P 500 index and the log of the S&P 500 index price
DY	Dividend yield, defined as the difference between the log of S&P 500 dividends and the log of lagged S&P 500 prices
SVAR	Stock return variance, calculated as the sum of squared daily returns on the S&P 500 index
BM	Book-to-market ratio, defined as the ratio of book value to market value for the Dow Jones Industrial Average
NTIS	Net equity expansion, calculated as the 12-month moving sums of net issues by stocks listed on NYSE divided by the total end-of-year market capitalization of NYSE stocks
TBL	Treasury bill rate, defined as the yield of a 3-month U.S. T-bill
LTY	Long-term yield, which is the long-term government bond yield.
LTR	Long-term return, defined as the return on long-term U.S. government bonds
TMS	Term spread, calculated as the long-term yield minus the U.S T-bill rate
DFY	Default yield spread, defined as the difference between BAA- and AAA-rated bond yields
DFR	Default return spread, calculated as the difference between the long-term corporate bond return and the long-term government bond return
INFL	Inflation, calculated based on the U.S. Consumer Price Index
ECON	The first principal component of 14 economic variables (i.e., earnings-price ratio, *EP*; dividend-payout ratio, *DE*; dividend-price ratio, *DP*; dividend yield, *DY*; stock return variance, *SVAR*; book-to-market ratio, *BM*; net equity expansion, *NTIS*; treasury bill rate, *TBL*; long-term yield, *LTY*; long-term return, *LTR*; term spread, *TMS*; default yield spread, *DFY*; default return spread, *DFR*; inflation, *INFL*)

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Appendix B: DFG in Equilibrium

In this appendix, we borrow the variance of variance model from Bollerslev et al. (2009) to explain the source of DFG and its return predictability. Suppose that the log growth rate of consumption is

$$g_{t+1} = \mu_g + \sigma_{g,t} z_{g,t+1}, \tag{A.1}$$

$$\sigma_{g,t+1}^2 = a_\sigma + \rho_\sigma \sigma_{g,t}^2 + \sqrt{q_t} z_{\sigma,t+1}, \tag{A.2}$$

$$q_{t+1} = a_q + \rho_q q_t + \varphi_q \sqrt{q_t} z_{q,t+1}, \tag{A.3}$$

where the details of the parameters could be referred to description for Equations (1)-(3) in Bollerslev et al. (2009).

Assuming the representative investor has Epstein-Zin-Weil recursive preferences, to save space, we just give the logarithm of the intertemporal marginal rate of substitution (IMRS) first to get the risk-free rate,

$$m_{t+1} = \log\delta - \psi^{-1}\mu_g - \frac{(\theta-1)(1-\gamma)^2}{2\theta}\sigma_{g,t}^2 + (\theta-1)\left(\kappa_1\rho_q - 1\right)A_q q_t - \gamma\sigma_{g,t} z_{g,t+1} + (\theta-1)\kappa_1\sqrt{q_t}\left[A_\sigma z_{\sigma,t+1} + A_q\varphi_q z_{q,t+1}\right], \tag{A.4}$$

where δ is the subjective discount factor, ψ is the intertemporal elasticity of substitution, γ is the coefficient of risk aversion, and $\theta = (1-\gamma)/(1-\psi^{-1})$.

Then the risk-free rate could be solved as

$$-rf_t = E_t[m_{t+1}] + \frac{1}{2}Var_t[m_{t+1}] = \log\delta - \psi^{-1}\mu_g - \frac{(\theta-1)(1-\gamma)^2}{2\theta}\sigma_{g,t}^2 + (\theta-1)\left(\kappa_1\rho_q - 1\right)A_q q_t + \frac{1}{2}\gamma^2\sigma_{g,t}^2 + \frac{1}{2}(\theta-1)^2\kappa_1^2 q_t\left(A_\sigma^2 + A_q^2\varphi_q^2\right). \tag{A.5}$$

We consider the one-period ahead risk-free rate as

$$rf_{t+1} = -\log\delta + \psi^{-1}\mu_g - \left[\frac{(1-\theta)(1-\gamma)^2}{2\theta} + \frac{1}{2}\gamma^2\right]\sigma_{g,t+1}^2 + \left[(1-\theta)\left(\kappa_1\rho_q - 1\right)A_q - \frac{1}{2}(1-\theta)^2\kappa_1^2\left(A_\sigma^2 + A_q^2\varphi_q^2\right)\right]q_{t+1}. \tag{A.6}$$

The conditional variance of the time t to $t + 1$ risk-free rate could be defined as $\sigma_{rf,t}^2 = Var_t(rf_{t+1})$, which could be solved as

$$\sigma_{rf,t}^2 = \left[\frac{(1-\theta)(1-\gamma)^2}{2\theta} + \frac{1}{2}\gamma^2\right]^2 q_t + \left[(1-\theta)\left(\kappa_1\rho_q - 1\right)A_q - \frac{1}{2}(1-\theta)^2\kappa_1^2\left(A_\sigma^2 + A_q^2\varphi_q^2\right)\right]^2 \varphi_q^2 q_t. \tag{A.7}$$

Similarly, we consider the one-period ahead variance of the risk-free rate as

$$\sigma_{rf,t+1}^2 = B_{rf} q_{t+1}, \tag{A.8}$$

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where $B_{rf} = \left[\frac{(1-\theta)(1-\gamma)^2}{2\theta} + \frac{1}{2}\gamma^2\right]^2 + \left[(1-\theta)(\kappa_1\rho_q - 1)A_q - \frac{1}{2}(1-\theta)^2\kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)\right]^2 \varphi_q^2$.

Given the IMRS, according to Drechsler and Yaron (2011), the variance risk premium in the risk-free rate could be calculated as

$$VRP_{rf,t} = E_t^Q[\sigma_{rf,t+1}^2] - E_t^P[\sigma_{rf,t+1}^2] = B_{rf}(\theta - 1)\kappa_1 A_q \varphi_q^2 q_t. \quad \text{(A.9)}$$

For the one-period ahead of the conditional variance of the market return, Bollerslev et al. (2009) show that

$$\sigma_{r,t+1}^2 = \sigma_{g,t+1}^2 + \kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)q_{t+1}, \quad \text{(A.10)}$$

and the variance risk premium in the market return

$$VRP_{r,t} = E_t^Q[\sigma_{r,t+1}^2] - E_t^P[\sigma_{r,t+1}^2] = (\theta - 1)\kappa_1[A_\sigma + \kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)A_q\varphi_q^2]q_t. \quad \text{(A.11)}$$

Now we define and decompose the DFG as

$$DFG_t = E_t^Q[\sigma_{rf,t+1}^2] - E_t^Q[\sigma_{r,t+1}^2] = (VRP_{rf,t} - VRP_{r,t}) + (E_t^P[\sigma_{rf,t+1}^2] - E_t^P[\sigma_{r,t+1}^2]), \quad \text{(A.12)}$$

where

$$VRP_{rf,t} - VRP_{r,t} = (\theta - 1)\kappa_1[B_{rf}A_q\varphi_q^2 - A_\sigma - \kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)A_q\varphi_q^2]q_t. \quad \text{(A.13)}$$

Please note that the objective conditional expectation of the variance of the risk-free rate could be obtained

$$E_t^P[\sigma_{rf,t+1}^2] = B_{rf}(a_q + \rho_q q_t), \quad \text{(A.14)}$$

and the objective conditional expectation of the variance of the market return also could be solved as

$$E_t^P[\sigma_{r,t+1}^2] = (a_\sigma + \rho_\sigma\sigma_{g,t}^2) + \kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)(a_q + \rho_q q_t). \quad \text{(A.16)}$$

Then the difference objective conditional expectations of two variances becomes

$$E_t^P[\sigma_{rf,t+1}^2] - E_t^P[\sigma_{r,t+1}^2] = -(a_\sigma + \rho_\sigma\sigma_{g,t}^2) + [B_{rf} - \kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)](a_q + \rho_q q_t). \quad \text{(A.17)}$$

Finally, we get closed-form solution for the DFG,

$$DFG_t = (\theta - 1)\kappa_1[B_{rf}A_q\varphi_q^2 - A_\sigma - \kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)A_q\varphi_q^2]q_t - (a_\sigma + \rho_\sigma\sigma_{g,t}^2) + [B_{rf} - \kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)](a_q + \rho_q q_t) = C_q q_t - \rho_\sigma\sigma_{g,t}^2 + [B_{rf} - \kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)]a_q - a_\sigma, \quad \text{(A.18)}$$

where $C_q = (\theta - 1)\kappa_1[B_{rf}A_q\varphi_q^2 - A_\sigma - \kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)A_q\varphi_q^2] + [B_{rf} - \kappa_1^2(A_\sigma^2 + A_q^2\varphi_q^2)]\rho_q$.

The model-implied equity risk premium (ERP) given by Bollerslev et al. (2009) is

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$$ERP_t = \gamma\sigma_{g,t}^2 + (1-\theta)\kappa_1^2\left(A_\sigma^2 + A_q^2\varphi_q^2\right)q_t. \quad \text{(A.19)}$$

The regression coefficient of DFG_t used to forecast market return is determined by

$$Cov(ERP_t, DFG_t) = C_q(1-\theta)\kappa_1^2\left(A_\sigma^2 + A_q^2\varphi_q^2\right)E[q_t^2] - \gamma\rho_\sigma E\left[\sigma_{g,t}^4\right]. \quad \text{(A.20)}$$

Given $\gamma > 1,\ \psi > 1$, we have $\theta < 0, A_q < 0, A_\sigma < 0$. In addition, given model parameters in Bollerslev et al. (2009), we have $C_q < 0$. This leads to

$$Cov(ERP_t, DFG_t) < 0. \quad \text{(A.21)}$$

Similarly, we also get

$$Cov\left(ERP_t, VRP_{r,t}\right) = -(1-\theta)^2\kappa_1^3\left[A_\sigma + \kappa_1^2\left(A_\sigma^2 + A_q^2\varphi_q^2\right)A_q\varphi_q^2\right]\left(A_\sigma^2 + A_q^2\varphi_q^2\right)E[q_t^2] > 0. \quad \text{(A.22)}$$

The negative covariation between DFG_t and the conditional ERP is mainly driven by $\theta < 0$; i.e., the investor prefers the early resolution of uncertainty. Similar to the VRP, the DFG is also a useful predictor for the market returns for which the volatility-of-volatility or q_t constitutes the dominant source of the variation in the ERP.

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Appendix C: Table A.1 Alternative measures for FTS and co-movement

Panel A: DFG with alternative measures for FTS and correlation

	(1) Ret(t)	(2) Ret(t)	(3) Ret[t+1, t+2]	(4) Ret[t+1, t+2]	(5) Ret(t)	(6) Ret(t)	(7) Ret[t+1, t+2]	(8) Ret[t+1, t+2]
$FTS^{alternative}$	-2.247***	-2.242***	0.041	0.035				
	(-39.70)	(-39.70)	(0.62)	(0.54)				
COR^{month}					-0.138***	-0.159***	-0.070*	-0.053
					(-3.04)	(-3.47)	(-1.80)	(-1.36)
DFG		0.126*		-0.222***		0.282***		-0.214***
		(1.65)		(-2.84)		(3.02)		(-2.71)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.3123	0.3127	0.0073	0.0120	0.0340	0.0373	0.0083	0.0125
N	4783	4783	4783	4783	4783	4783	4783	4783

Panel B: DFG with covariance between bond and stock market returns

	(1) Ret(t)	(2) Ret(t)	(3) Ret[t+1, t+2]	(4) Ret[t+1, t+2]	(5) Ret(t)	(6) Ret(t)	(7) Ret[t+1, t+2]	(8) Ret[t+1, t+2]
Covariance	-1.877***	-1.830***	-0.051	-0.098				
	(-7.59)	(-7.48)	(-0.24)	(-0.46)				
$Covariance^{month}$					-0.959***	-0.949***	0.006	-0.001
					(-4.39)	(-4.33)	(0.03)	(-0.01)
DFG		0.210**		-0.225***		0.244***		-0.223***
		(2.29)		(-2.90)		(2.61)		(-2.85)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.0500	0.0518	0.0073	0.0120	0.0458	0.0483	0.0073	0.0119
N	4783	4783	4783	4783	4783	4783	4783	4783

This table reports the results for additional tests considering flight-to-safety dummy and bond-stock market co-movement. $FTS^{alternative}$ is an alternative measure for flight-to-safety. It is a dummy equal to one if the daily stock market return is lower than -1% while the daily Treasury bond market return is greater than zero. $CORet^{onth}$ is an alternative measure for bond-stock market co-movement, calculated as one-month rolling correlation between daily bond and stock market

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returns. *Covariance* and $Covariance^{month}$ are other two alternative measure for bond-stock market co-movement, calculated as one-year and one-month rolling covariance between daily bond and stock market returns, respectively. Newey-West t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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Appendix D: Table A.2 Comparison with economic variables (monthly)

Panel A: Univariate regressions				Panel B: Bivariate regressions					
$Ret_{t+1} = \alpha + \varphi ECON_t^k + \varepsilon_{t+1}$				$Ret_{t+1} = \alpha + \beta DFG_t + \varphi ECON_t^k + \varepsilon_{t+1}$					
	φ	t-stat	Adj. R^2	β	t-stat	φ	t-stat	Adj. R^2	ΔR^2
DP	0.021	(0.72)	0.002	-5.000***	(-3.20)	0.033	(1.37)	0.074	0.072
DY	0.022	(0.84)	0.002	-4.946***	(-3.15)	0.033	(1.46)	0.073	0.071
EP	-0.004	(-0.40)	-0.003	-4.755***	(-3.01)	-0.007	(-0.70)	0.063	0.066
DE	0.005	(0.56)	-0.001	-4.876***	(-3.12)	0.009	(1.02)	0.068	0.069
SVAR	-0.041	(-0.04)	-0.004	-4.634***	(-2.85)	-0.048	(-0.06)	0.059	0.063
BM	0.104*	(1.87)	0.015	-5.013***	(-3.14)	0.130**	(2.35)	0.088	0.074
NTIS	0.266	(1.17)	0.007	-4.436***	(-2.91)	0.151	(0.77)	0.062	0.056
TBL	-0.168	(-1.14)	-0.001	-4.600***	(-2.74)	-0.028	(-0.19)	0.059	0.060
LTY	-0.445*	(-1.94)	0.012	-5.293***	(-2.57)	0.207	(0.76)	0.061	0.049
LTR	0.098	(0.90)	0.001	-4.570***	(-2.82)	0.074	(0.71)	0.062	0.061
TMS	-0.159	(-0.69)	-0.002	-4.960***	(-2.94)	0.165	(0.69)	0.061	0.063
DFY	-0.330	(-0.27)	-0.003	-4.756***	(-3.05)	0.288	(0.27)	0.060	0.063
DFR	0.097	(0.37)	-0.002	-4.703***	(-2.95)	0.131	(0.59)	0.063	0.065
INFL	0.403	(0.44)	-0.003	-4.633***	(-2.88)	0.401	(0.53)	0.060	0.063
ECON	0.002	(0.42)	-0.002	-5.402***	(-3.48)	0.005	(1.48)	0.077	0.079

This table reports the estimation results for predictive regressions on a single economic variable in Panel A, and bivariate predictive regressions that include both DFG and each economic variable in Panel B. The dependent variable is the one-month-ahead equity market return. *DFG* is defined as the residuals derived from regressing MOVE on VIX on a monthly basis. The descriptions of economic variables are available in Appendix A. Newey-West standard errors are estimated, and the corresponding t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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Appendix E: Table A.3 Comparison with other predictors (monthly)

	(1) Ret(t+1)	(2) Ret(t+1)	(3) Ret(t+1)	(4) Ret(t+1)	(5) Ret(t+1)	(6) Ret(t+1)
DFG	-5.236***	-5.405***	-6.027***	-5.285***	-5.494***	-5.732***
	(-3.59)	(-3.47)	(-3.76)	(-3.45)	(-3.32)	(-3.81)
Ret	-0.005					-0.006*
	(-1.41)					(-1.71)
FTS		-0.001				0.004
		(-0.07)				(0.49)
COR			0.029*			0.028*
			(1.86)			(1.90)
VRP				-0.014		-0.016
				(-1.00)		(-1.23)
RA					-0.002	-0.001
					(-0.32)	(-0.21)
Adj. R^2	0.0853	0.0725	0.0863	0.0833	0.0735	0.0971
N	228	228	228	228	228	228

This table reports the estimation results for monthly predictive regressions that include both *DFG* and other predictors. The dependent variable is the one-month-ahead equity market return. *DFG* is defined as the residuals derived from regressing MOVE on VIX on a monthly basis. *Ret* is the current monthly stock market return. FTS is a dummy equal to one if the current monthly stock market return is lower than -1% while the current monthly Treasury bond market return is greater than zero. *COR* is the one-year rolling correlation between daily Treasury bond and stock market returns. VRP is estimated as the difference between the physical and risk-neutral expectations of the future return variance. *RA* is the Risk Aversion Index (*RA*) constructed by Bekaert, Engstrom and Xu (2022), defined as the coefficient of relative risk aversion implied by the utility function and retrieved from the model and data on corporate bonds and equities. *ECON*, the first principal component of those14 economic variables, is included as a control. The descriptions of economic variables are available in Appendix A. Newey-West standard errors are estimated, and the corresponding t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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Appendix F: Table A.4 Extended sample period 1990 - 2022 (monthly)

Panel A

Forecast horizons	(1) m=1	(2) m=3	(3) m=6	(4) m=9	(5) m=12	(6) m=24
DFG	-3.170***	-2.617***	-2.486***	-2.218**	-1.863**	-0.397
	(-2.73)	(-3.23)	(-3.05)	(-2.49)	(-2.06)	(-0.54)
Adj.R^2	0.0271	0.0712	0.1372	0.1786	0.1914	0.1457
N	396	396	393	390	387	375

Panel B

	(1) Ret(t+1)	(2) Ret(t+1)	(3) Ret(t+1)	(4) Ret(t+1)	(5) Ret(t+1)	(6) Ret(t+1)
DFG	-2.979***	-3.167***	-3.832***	-3.099***	-3.166***	-3.759***
	(-2.71)	(-2.70)	(-2.94)	(-2.72)	(-2.70)	(-3.19)
Ret	-0.005**					-0.006**
	(-2.04)					(-2.01)
FTS		0.003				0.005
		(0.43)				(0.79)
COR			0.016**			0.022***
			(2.01)			(2.85)
VRP				-0.006		-0.015
				(-0.44)		(-1.18)
RA					0.002	0.006
					(0.36)	(1.10)
Adj. R^2	0.0398	0.0252	0.0369	0.0254	0.0256	0.0547
N	396	396	396	396	396	396

This table reports the estimation results for monthly univariate predictive regressions in Panel A and monthly multivariate regressions in Panel B, respectively, over an extended sample period from 1990 to 2022. *DFG* is defined as the residuals derived from regressing MOVE on VIX on a monthly basis. *Ret* is the current monthly stock market return. FTS is a dummy equal to one if the current monthly stock market return is lower than -1% while the current monthly Treasury bond market return is greater than zero. *COR* is the one-year rolling correlation between daily Treasury bond and stock market returns. VRP is estimated as the difference between the physical and risk-neutral expectations of the future return variance. *RA* is the Risk Aversion Index (*RA*) constructed by Bekaert, Engstrom and Xu (2022), defined as the coefficient of relative risk aversion implied by the utility function and retrieved from the model and data on corporate bonds and equities. *ECON*, the first principal component of those14 economic variables, is included as a control. The descriptions of economic variables are available in Appendix A. Newey-West standard errors are estimated, and the corresponding t-statistics are reported. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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